

IPSec Security Protocol

Module 14

Duration: 26:49

Video: https://www.youtube.com/watch?v=wVX5KLPh_o8

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Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar University of Science and technology Hisar harana today in this lecture we will study about IP SEC security protocol first of all please see the contents which I shall cover after covering brief introduction about IPC Internet Security protocols I shall take up its architecture which will be followed by IPC protocols ah that is authentication header and ESP that is encapsulating security payload in last I shall cover ik that is internet key exchange protocol for Key Management IP SEC Internet Protocol security is a framework for a set of protocols for security at the network or packet processing layer of network communication earlier security

approaches have inserted security at the application layer of the communication model IPC is set to be especially useful for implementing virtual private networks and for remote user access through dialup connection to PR private networks 2 minutes, 1 second a big advantage of IPC is that security Arrangements can be handled without requiring changes to individual user computers IPC is a protocol suit for securing Internet Protocol IP communication by authenticating and encrypting each IP packet of a communication session IPC can be used in protecting data flows between a pair of hosts means host to host between a pair of security gateways means Network to network or between a pair of security Gateway and a

host means Network to host IP SEC is is not a single protocol 3 minutes, 1 second but rather a set of services and protocols that provide a complete Security solution for an IP network these services and protocols combine to provide various types of protection some of the kinds of protection services offered by IPC include the following it offers confidentiality by encrypting data it provides Integrity by calculating checkm or hash value of the data and authentication by using signatures and certificates it provides protection against certain types of security attacks such as replay attack it also manages sessions along with key management now after covering basic introduction of Ip sec let us move to

its architecture RFC

241 defines the basic architecture of IP SEC it covers Concepts in which security Association sa provide the bundle of algorithms and data that provide the parameters necessary for operations of two IP protocols ah authentication header and ESP encapsulating security payload there are two modes in which these protocols work transport mode and tunnel mode in the end there is Key Management protocol like internet key exchange that is ik ke now let us first move on to modes of protocol in IP secc IPC can be implemented in a host to host transport mode as well as in a Network tunneling mode in transport mode only the payload of the IP packet is usually encrypted and authenticated the routing is intact because the IP header is neither

modified nor encrypted in tunnel mode the entire IP packet is encrypted or authenticated it is then encapsulated into a new IP packet with a new IP header other way to explain difference between transport mode and tunnel mode is that transport mode can only be used between end points of a communication or we can say transport mode is used when the cryptographic end points are also the communication end points of the secured IP packets figure shows transport mode in which communication and cryptographic end points are A and B tunnel mode can be used between arbitrary peers when at least one cryptography traic Endo is not 7 minutes, 1 second a communication Endo of the secured IP packets sa a indicates here security associations lower part of figure shows

tunnel mode in which communication takes place between either end point a and router R A or router r a a to router RB or router RB to endpoint B accordingly packet structure is shown in which source is router R A and destination is router RB tunnel mode is used to create virtual packet networks for Network to network communications host to network communications and host to host Communications IPC provides two choices of security service authentication header ah which essentially allows authentication of the sender of data and encapsulating security payload ESP which supports both authentication of the sender and encryption of data as well the specific information associated with each of these Services is inserted into the packet in a header that follows

the IP packet header now let us take each one by one authentication header ah is a member of the IPC protocol suit ah guarantees connectionless integrity and data origin authentication of Ip packets further it can optionally protect against replay attacks by using the sliding window technique and discarding old packets ah is one of the two core security protocols in IP SEC it provides authentication of either all or part of the contents of a datagram through the addition of a header that is calculated based on the values in the datag gr authentication header ah protocol work in both transport and tunneling mode figure shows packet format alteration with ah for Trans Port mode in this ah header is added after payload

that consist of TCP header and data next is figure which shows packet format alteration with ah for tunnel mode in this ah header is added after original Ip header and data here data also contains TCP or UDP header ah uses a simple algorithm which uses special hashing algorithm and a specific key known only to the source and the destination security Association sa between two devices specifies these particulars so that the source and destination know how to perform the computation but nobody else can on the source device ah performs the computation and puts result Integrity check value or icv into a special header with other fields for transmission the destination device does the same calculation using the key that the two devices

share this enables the device to see immediately if any of the fields in the original datagram were modified or not now let us move over to the format of authentication header please C figure which shows different fields of authentication header let us discuss each field one by one first field is next header it is of one byte which contains the protocol number of the next header after the a it is used to link headers together next field is payload length this field is of 1 byte and it measures the length of the authentication header itself including the payload the length of this authentication header is measured in two octet units minus 2 for example an ah value of 4 equal $3 + 3 - 2$ here first

three indicates first three rows of 32 bit each and next three indicates three rows for 32bit icv field and thus an ah value of 4 means 24 octets Third Field is reserve which is of 16 bits and is not used till now so it is set to zero next field is security parameter index SPI which is of 32 bits and it is arbitrary value which is used together with the destination IP address to identify the security Association of the receiving party security parameter index SPI is followed by sequence number field which is of 32 bits the field is incremented by one for every packet sent to prevent replay attacks when replay detection is enabled sequence numbers are never reused because a new security Association must

be renegotiated before an attempt to increment the sequence number Beyond its maximum value last field is integrity check value icv which is multiple of 32 bits this field may contain padding to align the field to an 8 octet boundary for IPv6 or a four octet boundary for ipv4 the IPC authentication header ah provides Integrity authentication services to IP SEC capable devices so that they can verify that messages are received intact from other devices for many applications however this is only one piece of the puzzle we want to not only protect against intermediate devices changing the datagram but also to protect against them examining their contents as well for this level of private communication

ah is not enough we need to use the ESP protocol because ah just provides authentication but not privacy privacy is provided by es now let us see next protocol in IPC that is encapsulating security payload that is es encapsulating security payload ESP is a member of the IPC protocol suit IPC provides origin authenticity Integrity confidentiality and protection of packets ESP operates directly on top of Ip the main job of ESP is to provide the Privacy we seek for IP

datagram by encrypting them an encryption algorithm combines the data in the datagram with a key to transform it into an encrypted form ESP can also be implemented in both transport and tunneling mode figure shows packet format alteration with ESP for transport mode

in this ESP header is added after payload that consist of TCP header and and data ESP trailer and ESP authentication fields are added before TCP header and data however in tunneling mode where the entire original Ip packet is encapsulated with a new packet header added ESP protection is afforded to the whole inner IP packet while the outer header remains unprotected as in transport mode ESP trailers and ESP authentication are added before data now let us move over to the format of encapsulating security payload ESP please see the figure which shows different fields of ESP let us discuss each field One By One ESP has several fields that are same as those used in ah but it packages its field in a very different way instead of

having just a header it divides its Fields into three components first is ESP header this contains two Fields SPI and sequence number and they come before the encrypted data its placement depend on whether ESP is used in transport mode or tunnel mode as explained earlier in my lecture next is ESP trailer this section is placed after the encrypted data it contains padding that is used to align the encrypted data through a padding and Pad length field interestingly it also contains the next header field for ESP ESP authentication data this field contains an icv that is computed in such a manner that is similar to how the ah protocol works the field is used when esp's optional authentication feature is employed now let us see each of these

field one by one first field is security parameter index which is of 32 bits it stores an arbitrary value used together with a destination I I address to identify the security Association of the receiving party next field is sequence number which is also of 32 bits it is monotonically increasing sequence number which increments by one for every packet sent to protect against replay attacks there is a separate counter kept for every security Association Next we move on to payload data field which is variable after payload data it is padding which is of 0 to 255 octets it is used as padding for encryption to extend the payload data to a size that fits the encryption Cipher block size and to align the next field next field is pad length field which is

of 8 Bits And it specifies size of padding in octets moving on to next header field which is of 8 Bits And contains next Header information now there are two reasons why these fields are broken into pieces like this the first is that some encryption algorithms require the data to be encrypted to have a certain block size and so padding must appear after the data and not before it that's why padding appears in the ESP trailer field the second is that the ESP authentication data appears separately because it is used to authenticate the rest of the encrypted datagram after encryption this means that it cannot appear in the ESP header or ESP trailer next we shall discuss Key Management protocol in IP SEC which is internet key

exchange. Like many secure networking protocol sets, it is based on the concept of a 23-minute, 1-second shared secret between two devices that want to send information securely and encode and decode it using a piece of information that only the devices know. Anyone who isn't on the secret is able to intercept the information but is prevented either from reading it or from tempering it. The primary support protocol used for this purpose in IPsec is known as Internet Key Exchange (IKE). That is, IKE is defined in RFC 2409 and it is more complicated of the IPsec protocols to comprehend. In fact, it is simply impossible to truly understand more than a real simplification of its operation without significant background in

cryptography. The purpose of IKE is to allow devices to exchange information that is required for secure communication, as the title suggests. This includes cryptographic keys that are used for encoding authentication information and performing payload encryption. IKE works by allowing IPsec-capable devices to exchange SAS. Main features of IKE are as shown in Figure 1. IKE is an IPsec component used for performing Mutual authentication and establishing and maintaining security associations. It is typically used for establishing IPsec sessions. It has a key exchange mechanism for security at Network layer. It has five variations of an IKE negotiation with two modes: aggressive and Main. It also has three authentication

methods: pre-shared, public key encryption, and public key signature. So dear friends, today in this lecture, we have covered introductory concepts of IPsec protocol followed by its architecture. Then we covered IPsec protocols: AH that is authentication header and ESP that is encapsulating security payload. In the last, IKE that is Internet Key Exchange protocol for Key Management was covered. Thank you.